

Mental Health Prognosis

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Abstract— Mental health is becoming a major problem in this decade with over 60%, and rising, of college students having at least one mental health problem[1]. Using machine learning we can develop classifier models to help identify those in need or in danger of developing a disorder.

Index Terms— mental health, models, classifiers, datasets, KNIME, decision tree, random forest, ensemble, overfitting, correlation, neural network, one-hot-encoding, normalization

I. Introduction

The rise in mental health disorders specifically among college students has become a problem, not just for the students but for counselors as well. Many universities have tried to combat this by tripling their staff, but it can only do so much as many students still go unchecked with mental health disorders[1]. So how can we combat this? Many datasets on kaggle.com have already shown many of the attributes students with these conditions possess[4]. So utilizing machine learning technology (ie. classifiers) we can predict, the best we can, students who might need help and are not reaching out. Stabler shows in her article that mental health disorders are one of the leading causes of suicide in college students[5]. Giving counselors the resources to be able to classify students at risk would only help to reduce many problems students face.

II. Decision Trees

Decision trees are useful for categorizing results where the attributes (age, college major, gpa, etc) can be sorted against known datasets[3]. The main type of decision tree I am going to use is a categorical tree where the answer I want to predict is “do they have depression/anxiety?” Regarding mastersindatascience.org they state categorical trees are best for when “... the answer neatly fits into one category or another.” [3] Knime has straightforward documentation on how to implement these (i.e Decision Tree Learner) [2]. Using an open source dataset from Kaggle.com by Shariful Islam [4] I developed a decision

tree classifier for predicting whether a student could have depression and/or anxiety based on a variety of attributes ranging from age, major, and marital status. While developing this KNIME workbook I found out there is an almost direct correlation between marital status and depression and/or anxiety. Where every college student who was married during school had depression. After messing around with the decision tree learners node options I was able to get my accuracy from 63% to around 82% mainly from changing the quality measure from “Gini index” to “Gain ratio”. Even if the model wrongly predicts a student who does not have depression they can still be at risk of developing it later. Like with regular medical conditions it is not an absolute science. As for predicting anxiety, changing the learner’s quality measure from “Gini index” to “Gain ratio” actually decreased accuracy. This tells me there is no direct correlation between depression and anxiety. Those with depression are more likely to have anxiety but those with anxiety are not necessarily more likely to develop depression. Regarding my model for predicting anxiety. The model scored an average accuracy around 65% and no node options really increased this. This tells me two things: one the attributes used don't necessarily have a relationship to someone with anxiety, two the model I used is not very suitable to this specific dataset for that prediction category. Overall, even though the decision tree learner was not super accurate it gives great insight into what attributes most directly correlate to those with mental health conditions.

III. Random Forest Learner

An ensemble of decision trees is called a random forest learner. Using multiple decision trees to classify data. Each of the trees in the forest is trained on a random subset of the training dataset then the final prediction is made by combining all these trees [6]. It can lead to higher accuracy than a single decision tree. These learners are more resilient to overfitting compared to single decision trees. Though these learners are more computationally heavy than decision trees the excess power needed is compensated for the increased accuracy. Now how does this directly compare to decision trees on the same dataset I used earlier. Regarding predicting depression in students

both models seemed to have around the same accuracy, but in the random forest when I excluded the major attribute the model accuracy jumped up to 85%. Now predicting anxiety with the same datasets the random forest learner was around 10% more accurate but the accuracy is still very low at around 62%. Changing options within the random forest learner node seemed to change any accuracy meaningfully. This result just reinforces my earlier conclusion that this dataset is not well suited to classify those with anxiety. The attributes used do not have any sort of correlation with students who have anxiety.

IV. TensorFlow Neural Network

After the KNIME implementation I wanted to see if using a neural network would be more accurate. Using python, tensorflow, pandas, and sklearn I developed a model with 3 layers, 1 flatten layer, and 3 dense layers. The first layer does not have any trainable parameters, just flattens the dataset so that it can be processed by the other layers. The issue with developing a neural network using the same dataset is that many of the categories are string values, so I have to one-hot-encode each of those categorical columns into numerical values. Pandas has a simple way to do this with the `get_dummies()` function that converts values into "...as many 0/1 variables as there are different values"[7]. This was helpful but created many many redundant columns that had to be removed for computational sake. Then using sklearn's `train_test_split()` function I split that data into 85% training and 15% test set. After that I had to normalize that data based on 255. I do this so that all the variables/values are on a similar scale. Since if age and GPA were not a part of the dataset normalization would not necessarily be needed though still might have a slight impact on accuracy. After I define the neural network architecture. After some experimentation adding an extra 4th, 5th and even 6th layer proved to have no impact on the overall accuracy. Next I had to compile the model using an 'adam' optimizer, the loss function 'sparse_categorical_crossentropy' and for evaluation metric an 'accuracy' was used. Then we have to fit the model and iterate over the data a set number of times, epochs. After some trial and error epochs and the amount of layers have no real effect on the accuracy of the model. My model scored a 68.75% accuracy on predicting depression which is much lower than the decision tree and random forest learners I used in KNIME. After some test and training data modification, I saw that a 90% training and a 10% test resulted in a 72.72% accuracy.

V. Challenges

One of the challenges I faced was validating datasets for a neural network. Specifically using a pandas dataframe and manipulating categorical data and one-hot encoding them into numerical values. This process can be tedious if done by hand so I looked into pandas documentation and tried to validate my dataset. When trying to one-hot-encode major category using the `get_dummies()` function of pandas it resulted in adding around 40+ extra columns, though this was not a huge problem it can be more computationally heavy than if I were to one-hot-encode all majors into one category. I could not figure out how to do this without manually editing the csv file.

VI. Conclusion

After a lot of experimentation I made some interesting conclusions about relationships between attributes in the dataset. The first major one is that there is a direct relationship between marital status and depression, another is anxiety does not seem to fit well with these specific attributes. I believe that if the dataset I found on kaggle was bigger than 100 students my accuracy would increase. Overall, mental health prognosis is not an absolute science and there are always outliers, but something I did find super interesting was that people with depression are very likely to have anxiety but those with anxiety are not always likely to have depression. This is something I just found out after analyzing the decision tree results. Even if results are false-positives, therapists can still help prevent depression/anxiety before a student gets it. If we take that into account the accuracy of the models is theoretically higher. Since we want to help those who are suffering and those at risk compared to others.

References

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Code Repository

<https://github.com/notfennecks/MentalHealthMachineLearning>